

Certified Recursive Bisection: Proof-Carrying Cuts for Algorithmic Redistricting

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Abstract

Recursive graph bisection offers an intelligible construction rule for redistricting, but practical partitioners ordinarily return heuristic cuts without proving that a better permitted cut does not exist. It therefore remains unresolved whether a fixed bisection schedule can support both model-scoped optimality evidence and staged public scrutiny without implying that its governing objective is uniquely fair. We present *certified recursive bisection*, a proof-carrying architecture that separates institutional choice of a rule from verifiable execution. The procedure freezes the unit universe, population, adjacency, district-count tree, lexicographic objective, and canonical tie-break; reconstructs child instances from certified parent assignments; and binds all internal cuts to one-seat leaves. A public ceremony commits the rule before disclosure and releases certified cuts by tree depth after fixed review intervals. On eight precommitted synthetic graphs and five METIS seeds, heuristic cuts match the exact primary objective in 30 of 40 rows and the exact canonical assignment in 20. In a four-district ceremony fixture, all three internal cuts are released across two rounds, the final tree binds, and a timestamp mutation is rejected. Rhode Island demonstrates a 25,649-block population-proof frontier. A clean-worktree replay exactly regenerates all 150 State-cycle assignments across three national operational baselines, whose 1,155 recursive nodes are within tolerance but provide zero State-scale weighted-boundary or canonical proofs. These results supply a concrete evaluation contract: future bakeoffs can report map outcomes and procedural evidence separately, while legislatures and commissions retain responsibility for legal, representational, and normative choices.

1 Introduction

Congressional redistricting combines a fixed numerical obligation—create exactly k districts—with a vast collection of geographically feasible assignments. Recursive bisection reduces that choice space to a sequence of two-way decisions. For California’s 52-district delegation, the canonical schedule begins

$$52 \rightarrow 26/26, \quad 26 \rightarrow 13/13, \quad 13 \rightarrow 6/7.$$

The schedule is intelligible, reproducible, and naturally parallel. The remaining discretion lies in selecting each cut.

Practical graph partitioners such as METIS are well suited to finding strong balanced cuts quickly [4]. Their output, however, is not an optimality proof. Re-running a deterministic heuristic can establish what the program produced; it cannot establish that no better cut was permitted by the same rules.

This paper asks whether the recursive structure itself can become proof-carrying. Our answer is conditional but constructive. A governing rule must freeze the unit universe, population, adjacency and island links, split schedule, objective order, and tie-break. Software can then certify each required cut and bind the cuts into one recursive tree. This follows the procedural lesson of Huntington–Hill: mathematics settles execution only after the governing criterion has been selected [3].

The contribution is a methods and evidence contract. We do not claim that the chosen geographic objective is inevitable, that the resulting plan is a legal safe harbor, or that certified plans already outperform all alternatives on compactness, representation, or runtime.

2 Fixed Recursive Model

For a parent region requiring $k \geq 2$ districts, define

$$k_L = \left\lfloor \frac{k}{2} \right\rfloor, \quad k_R = k - k_L.$$

Every unit is assigned to exactly one child, both children are nonempty and connected, and each child contains at least as many units as eventual districts. The solver may choose the cut; it may not alter (k_L, k_R) .

Let P be parent population and P_L left-child population. The seat-ratio-scaled deviation is

$$\Delta_L = |kP_L - k_LP|.$$

The right-child expression is algebraically equal. We retain maximum and total deviation fields for schema continuity, although total deviation is twice the maximum for a binary split.

Among feasible assignments, the procedure minimizes lexicographically:

- (1) maximum scaled population deviation;
- (2) total scaled population deviation;
- (3) weighted boundary cut; and
- (4) the canonical binary assignment.

Equal-seat splits have label symmetry, removed by placing canonical unit zero in the left child. Unequal-seat splits are already oriented by the floor/ceiling seat counts; forcing unit zero left could exclude the ratio-correct solution and is therefore prohibited.

Island components use the project’s deterministic rule: each non-main- component unit connects to its nearest same-county unit in the main component, with statewide fallback. Synthetic bridge edges receive the median land-boundary weight.

3 Split And Tree Certificates

A split instance binds its tree path, parent-certificate identity, canonical unit universe, populations, weighted edges, seat counts, and orientation rule. The result is either an optimal connected assignment or exact infeasibility. The bounded reference oracle commits to every candidate in deterministic order.

A whole-plan certificate must establish more than the validity of individual cuts. The verifier reconstructs both child contexts from each certified parent assignment. Submitted child populations, edges, and unit IDs are not trusted. Non-leaf nodes must appear in the breadth-first order of the canonical bisection schedule. One-seat leaves are ordered by binary path and must partition the root universe exactly once.

This construction distinguishes algorithmic proof from plan audit. An RPLAN audit can confirm assignment shape, population, contiguity, and provenance. The recursive certificate establishes the stronger model-scoped claim that no lexicographically better cut exists at each certified node.

The guarantee remains sequential. If a locally optimal parent cut produces a child that cannot be split under the same rules, the procedure fails explicitly; it does not silently backtrack to a worse parent cut.

4 A Public Sequential-Cut Ceremony

A certificate tree proves a completed computational object. It does not, by itself, show when officials committed to the rule or whether intermediate cuts were disclosed before later choices could be substituted. We therefore define a ceremony around the same canonical tree.

Before geographic cuts are released, a genesis transcript commits the rule- profile hash, root-instance hash, district count, floor/ceiling tree, minimum review interval, witness threshold, and challenge policy. For a plan with k districts, the completed binary tree contains exactly $k - 1$ non-leaf cut decisions. Round zero releases the root. Round d releases every non-leaf cut at depth d , ordered by binary path, only after the frozen interval has elapsed since the prior publication.

Every announcement embeds its complete certified split instance, certificate, and proof. The round also binds its timestamp, external witness receipts, and the prior round hash. An observer can consequently verify the public prefix, including exact parent-derived child universes, without receiving unreleased descendants. The final round changes the transcript to **completed** and binds the complete certified-tree identifier. A challenged process may instead become permanently **halted**; its accepted prefix and nonempty halt reason remain verifiable, and resumption requires a new ceremony identifier.

Depth batching is deliberate. Requiring a separate statutory delay for all $k - 1$ cuts would make a large delegation needlessly serial, while disclosing all cuts at once would eliminate intermediate review. Same-depth cuts depend on already accepted parents and are mutually independent, so they may be reviewed together. Different States may likewise proceed in parallel.

The reference implementation exposes immutable **start**, **announce**, **verify**, and **halt**

transitions. It checks minimum elapsed time and distinct receipt bindings, but opaque receipt strings are not proof of real-world publication. An official deployment must validate the signatures and clocks of its named transparency logs or timestamp authorities.

This ceremony sharpens the Huntington–Hill analogy. The political institution selects and publishes the governing rule; each subsequent geographic decision is a locally checkable consequence of that rule. It does not make the rule value-free, prove a partisan-symmetric outcome, or displace legal and public review.

For prospective bakeoffs, ceremony evidence forms a separate procedural scoreboard: proof coverage, prefix verification, release completion, transcript size, verifier time, halt behavior, and hostile-test resistance. These fields must not be combined with map-quality metrics into a post hoc scalar winner. The preexisting bakeoff results predate this protocol and are not retroactively classified as ceremony runs.

5 From Discovery To Proof

Scalable certification separates two responsibilities. A fast discovery solver proposes a connected assignment and exact objective tuple. That record is an incumbent, not a proof.

Certification compiles three pseudo-Boolean decision problems:

1. Is any feasible cut better on population?
2. At the accepted population bound, is any cut better on boundary?
3. At both bounds, is any canonical assignment lexicographically earlier?

A satisfiable decision returns a counterexample and rejects the incumbent. Three independently checked unsatisfiability proofs establish the unique cut. The prototype emits OPB models intended for a proof-producing solver such as RoundingSat and an independent checker such as VeriPB [2, 1].

The initial compiler used static connectivity no-goods generated by bounded enumeration. The implemented `parent-depth-v1` alternative instead selects one root per child, one parent arc for each assigned non-root unit, and binary depths that increase strictly along selected arcs. This polynomial-size encoding has a committed RoundingSat boundary proof accepted by VeriPB.

The static compiler remains exponential in the number of units. When the incumbent already attains zero population deviation, the first lower-bound query is a trivial contradiction; the boundary and canonical stages remain the substantive optimality decisions.

6 Implemented Evidence

The bounded corpus establishes contract correctness, identity binding, and tamper resistance. The exact reference oracle remains bounded, but the operational certified-discovery path now establishes State-scale execution, complete block coverage, connected recursion, and independent package verification. These are distinct evidence layers: operational verification does not convert a discovered incumbent into an exact objective proof.

The ceremony tests establish the software transition contract, including early round rejection, duplicate-witness rejection, child-substitution rejection, immutable output paths, and verification of an open prefix without disclosure of the future tree. They do not establish that an external witness actually observed a publication at the asserted time. No historical national or structure-family bakeoff has yet been rerun under an externally witnessed ceremony; comparative externally witnessed results remain pending.

A bounded path-8 procedural bakeoff now exercises the reporting contract. For $k = 4$, the certified arm reports three of three certified cuts, two accepted depth releases, prefix verification, completed status, and final-tree binding. A matched METIS replay arm remains in the same matrix as **replay-only**, with zero proof coverage and unsupported ceremony status. The independent regenerator reproduces every deterministic field and rejects a mutated round timestamp. Receipt strings are test values, external witnesses are not validated, and the full frozen negative corpus is not claimed as run.

The State-scale frontier now goes one step further. Deterministic METIS plus nine articulation-safe boundary moves reaches Rhode Island populations 548,689 and 548,690. RoundingSat proves that deviation below 1 is impossible, and VeriPB accepts that proof. The boundary decision timed out, so the root cut is not yet fully certified.

Across the three national cycles, all 1,155 recursive nodes satisfy the frozen population tolerance. Thirteen nodes attain the arithmetic lower floor. Exact weighted-boundary and canonical proof coverage remains 0 of 1,155. The structural comparison finds 120 node signatures common to all three cycles and 18 States with exact all-cycle tree topology. Pair-wise topology Jaccard is 0.328 for 2000–2010, 0.283 for 2000–2020, and 0.360 for 2010–2020. Cross-census assignment overlap is deliberately uncomputed because changing Census block universes require a relationship crosswalk or geometry overlay.

The strongest demonstrated comparative result therefore remains epistemic: certified BI-SECT can state what would count as proof, distinguish SAT counterexamples from UNSAT certification requests, bind the final plan to the recursive evidence tree, and separate national operational completion from exact-objective coverage.

The governed 2020 plan-family bakeoff adds a descriptive national comparison without changing that proof boundary. Tier 1 retains 7,889,194 land-containing blocks from 8,126,956 source blocks under one frozen water rule. After State-level maximum-overlap label matching, 4,194,107 block assignments match and 3,695,087 differ, a block-weighted agreement rate of 53.162680 percent. NRS splits 1,808 county units and 19,789 tract units; the official CD118 block projection splits 404 and 4,720, respectively.

Tier 2 dissolves those same blocks into common plan geometry. The district-weighted NRS and comparator means are 0.025380339 and 0.061087819 for Polsby–Popper, 0.245552262 and 0.377238283 for exact Reock, 0.557042985 and 0.702220801 for convex-hull ratio, and 7.847347095 and 5.656288737 for Schwartzberg. State-weighted means have the same direction. A 2026-09-09 clean-detached-worktree replay from the separately held 6.7 GiB certified context corpus regenerated all 150 State-cycle assignments, matched every governed assignment hash, and passed both 50-State Tier 1 regeneration and the 50-State Tier 2 geometry regeneration. The corpus and large regenerated packages remain outside Git, so a new machine still requires the documented data transfer. These are block-projected measurements rather than original enacted polygon linework. Their aligned direction does not establish a winner, fairness, legal validity, or superiority. Elections, demographics, non-enacted com-

parators, and converged block-level ensembles remain separately gated.

The precommitted comparison expands the original path-8 demonstration to eight bounded synthetic graphs and five fixed METIS seeds per graph. All 40 heuristic rows are feasible and connected. METIS matches the complete exact primary objective in 30 rows and the exact canonical assignment in 20. Path-8, the varied 10-cycle, the barbell, and the unequal tree match both assignment and objective for all seeds. The weighted grid and 20-cycle match the objective but select different tied assignments.

The ten objective disagreements are concentrated in two fixtures. On the varied ladder, METIS selects scaled population deviations (20, 40) and boundary 2 while the exact certified objective selects (0, 0) and boundary 4. On the unequal 4×4 grid, METIS selects (95, 190) and boundary 5 while the exact result selects (40, 80) and boundary 8. These are not contradictions: the frozen lexicographic model minimizes population deviation before boundary. The independent Python verifier reconstructs the eight instances and hashes, recomputes objectives, connectivity, row status, and all aggregates, and binds the report to the precommitted protocol. The finite suite demonstrates both agreement and meaningful disagreement; it cannot support State-scale, national map-quality, or runtime superiority.

7 What Certification Establishes

If all proof stages are independently accepted, certification establishes:

- the enacted recursive schedule was followed;
- every parent and child used the declared unit universe;
- no permitted connected cut improved the objective at any node;
- the final assignment is the unique canonical BISECT plan; and
- when the ceremony also verifies, accepted cuts were disclosed in canonical depth order with the frozen minimum intervals and hash chain.

Certification does not establish that the enacted axioms are normatively correct. It does not optimize partisan symmetry, competitiveness, communities of interest, racial representation, or every compactness measure. VRA and constitutional application remain separate legal questions. Nor does a self-asserted timestamp prove public availability: the production ceremony claim additionally requires validation by external witness systems.

Nor does the present implementation show that certified recursion “wins” on county preservation or nationwide geometry. The 2020 national bakeoff supplies a common-block descriptive comparison with official CD118 assignments, but it does not measure original enacted linework and does not turn four metric directions into a composite quality ordering. The precommitted eight-instance suite now runs the same bounded root instances through exact certified and heuristic selection without choosing seeds after the result. It is a comparison-mechanics result, not a representative political or State-scale sample. A defensible superiority study must run the same instances through certified and heuristic cut

selection, report objective agreement and disagreement, measure runtime and proof size, and evaluate final plan metrics without selecting favorable states or seeds.

The current conclusion is narrower: relative to heuristic output, solver metadata, and RPLAN audit alone, certified recursive bisection provides the strongest model-optimality claim architecture currently implemented in the project. The bounded comparison shows why proof can matter even when every heuristic result is feasible: ten of 40 rows sacrifice the model’s first-priority population objective for a smaller boundary. NRS v0.3 demonstrates that the operational recursion, assignment, and verification layer can run nationally across three Census cycles. Whether the boundary and canonical proof strength can be delivered at that scale remains the central open engineering question. National election, demographic, alternative-structure, and converged-ensemble comparisons remain open empirical questions under separate input and preregistration gates.

Future bakeoffs therefore report two non-combinable result families. The first retains map validity, objective values, downstream metrics, runtime, and timeouts. The second reports procedure class, certified-cut coverage, prefix verification, completion or halt status, transcript size, verifier time, and negative-corpus results. This avoids declaring an overall winner by choosing weights after observing a tradeoff between map outcomes and proof strength.

8 Conclusion

Certified recursive bisection preserves the most recognizable feature of BISECT—a fixed sequence of proportional two-way cuts—while defining how unproved per-node choices are replaced by a unique, checkable result. Bounded implementation, hostile fixtures, package verification, Rhode Island’s exact population frontier, a precommitted multi-instance heuristic comparison, and three complete national operational baselines show that the architecture is concrete rather than aspirational.

The implemented sequential-cut ceremony makes that architecture publicly staged: the rule is committed before the root, each later depth waits for a review window, observers can verify disclosed prefixes without future cuts, and a challenge can preserve a permanently halted record. The bounded path-8 fixture exercises a ceremony-enabled procedural scoreboard, but externally witnessed State ceremonies and State-scale ceremony-enabled bakeoffs remain future evidence rather than completed empirical claims.

The remaining work is equally concrete: integrate the compact encoding and pinned proof toolchain at State scale, integrate an exact discovery solver, obtain the first independently checked unrestricted State boundary proof, and extend the completed common-block geometry bakeoff to separately governed election, demographic, alternative-structure, and converged-ensemble comparisons. Until then, METIS-backed certified discovery remains the practical nationwide engine and exact certified BISECT remains the proof-oriented North Star.

References

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Artifact	Current evidence
Bounded split oracle	Optimal and infeasible certificates for equal and unequal seat ratios, with transcript tamper rejection.
Recursive tree	Parent-derived child contexts, canonical breadth-first nodes, and exact one-seat leaf coverage.
Public ceremony	Hash-chained genesis and depth-round transcripts, immutable CLI transitions, prefix-only verification, review windows, witness thresholds, completion binding, and permanent halt behavior.
CLI package	Tree, RPLAN, RCTX, audit certificate, and manifest with tree-to-plan verification.
Hostile evidence	False optimum, ID tamper, node-order, missing-leaf, and leaf-universe substitutions are rejected.
Precommitted comparison	Eight bounded synthetic graphs, five fixed METIS seeds per graph, and all 40 heuristic outcomes retained and independently recomputed.
Proof prototype	Static and compact OPB requests, a deliberate suboptimal SAT counterexample, and pinned RoundingSat/VeriPB population and compact-boundary proofs.
Rhode Island input	Connected 25,649-block RCTX with 66,097 land edges and 64 deterministic island bridges.
National operations	NRS v0.3 wall-to-wall block assignments for 50 States, 435 districts, and 385 recursive nodes in each of 2000, 2010, and 2020, with independent population and connectivity verification.
Cross-cycle evidence	Hash-bound node snapshots and a separately checkable three-cycle topology and descriptive incumbent-cut matrix.
National plan-family bakeoff	Published precommitted 2020 common-block comparison with official CD118: 50 States, 435 districts per family, Tier 1 assignment and subdivision metrics, and Tier 2 block-projected geometry. A clean-worktree vault-backed replay re-generated all referenced assignments.

Table 1: Implemented evidence and its present claim boundary.